

Q-AeroSolve

A Hybrid AI-Quantum Framework for Predictive Aerodynamic Modeling and CFD Acceleration

A Scalable Engineering Decision System for Spectral CFD, Tensor Compression, AI-Augmented Flow Prediction, QUBO Optimization, Quantum-Compatible PDE Evolution, and Distributed Quantum Simulation using Fujitsu Quantum Simulator

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Participant Name: J K Pawan Kumar and Tarun Solanki

User ID: g155-user1 and g155-user2

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Executive Summary

Problem

High-fidelity aerodynamic simulation is computationally expensive due to the non-linear and high-dimensional structure of the incompressible Navier–Stokes equations. As Reynolds number, spatial resolution, and simulation horizon increase, conventional CFD workflows face rapidly growing runtime and memory costs, limiting scalability for aerospace optimization and large-scale scientific computing.

Innovation

Q-AeroSolve introduces a hybrid AI–quantum computational framework integrating:

- Validated Computational Fluid Dynamics (CFD)
- FFT-based spectral PDE acceleration
- Tensor-based low-rank compression
- Machine learning error correction
- Quantum-compatible amplitude encoding
- Distributed mpiQulacs simulation using Fujitsu QARP

The framework establishes a scalable pathway connecting classical aerodynamic simulation, AI-enhanced numerical acceleration, and future quantum-compatible scientific computing systems.

Key Results

Metric	Result
Spectral Runtime Acceleration	24.8× faster than CFD baseline
Spectral Error Reduction	>90% reduction in numerical error
Tensor Compression Ratio	15.9× memory reduction
Tensor Reconstruction Error	Negligible reconstruction loss ($< 10^{-15}$)
ML Error Reduction	>98% prediction error reduction
Divergence Preservation	$\mathcal{O}(10^{-5})$ stability maintained
Quantum-Compatible Scaling	Distributed amplitude-state encoding validated
Fujitsu QARP Integration	mpiQulacs multi-rank execution validated

Why Hybrid Wins

No single computational strategy solves all aerodynamic scaling challenges efficiently. Spectral methods accelerate PDE evolution, tensor decomposition reduces memory complexity, machine learning stabilizes long-horizon prediction, and quantum-compatible encoding enables scalable state representation. Q-AeroSolve combines these complementary strengths within a unified computational workflow.

Why Fujitsu QARP Matters

Distributed mpiQulacs execution enabled:

- Multi-rank quantum-compatible state-vector simulation
- Scalable amplitude encoding of aerodynamic CFD fields
- Investigation of hybrid HPC–quantum scientific workflows
- Distributed execution for high-dimensional PDE systems

The Fujitsu Quantum Simulator environment therefore served as a practical platform for exploring next-generation quantum-compatible scientific computing architectures beyond traditional optimization-focused workloads.

Business Impact

- Faster aerodynamic optimization workflows
- Reduced simulation runtime and memory cost
- Scalable infrastructure for digital engineering systems
- Future-ready hybrid HPC–AI–quantum simulation architecture

Final Takeaway

Q-AeroSolve demonstrates that hybrid AI–quantum computational architectures can substantially improve aerodynamic simulation efficiency while preserving physical correctness and numerical stability. More importantly, the framework establishes a scalable intermediate pathway connecting validated CFD infrastructure, AI-enhanced numerical acceleration, and future quantum-compatible scientific simulation systems for aerospace and high-performance scientific computing applications.

1. Abstract

High-fidelity aerodynamic simulation governed by the Navier–Stokes equations remains computationally expensive due to the non-linear and high-dimensional nature of turbulent fluid dynamics [1, 3]. As Reynolds number, spatial resolution, and simulation horizon increase, conventional Computational Fluid Dynamics (CFD) workflows experience rapidly growing runtime and memory requirements, limiting scalability for large-scale aerospace optimization and scientific computing applications [2].

This work presents **Q-AeroSolve**, integrating validated CFD, spectral acceleration, tensor compression, AI forecasting, and distributed quantum-compatible PDE-state evolution using Fujitsu Quantum Simulator. Taylor–Green vortex benchmarks evaluate Reynolds regimes from $Re = 10$ to $Re = 1000$ across grid resolutions up to 256^2 cells [8].

Experimental results demonstrate substantial scalability improvements while preserving physical flow evolution. Spectral acceleration achieved **24.8** $\times$ runtime speedup with **>90%** error reduction. Tensor compression delivered **15.9** $\times$ memory reduction, while quantum encoding achieved **3.97** $\times$ latent-state compression for 12-qubit amplitude encoding. AI forecasting attained relative error of 6.32×10^{-16} under reduced-order benchmark conditions [5]. The framework extended to stable 3D 32^3 vortex evolution.

A major contribution of this work is the implementation of distributed quantum-compatible PDE-state evolution using mpiQulacs within the Fujitsu Quantum Simulator environment [10]. Multi-rank quantum evolution experiments were successfully executed across 6–12 qubits with scalable state dimensions up to 4096 amplitude states under MPI-based parallel execution, establishing a practical hybrid tensor–quantum pathway for scalable aerodynamic field representation.

Q-AeroSolve does not claim near-term quantum replacement of industrial CFD. Instead, it demonstrates a validated hybrid pathway where spectral solvers, tensor compression, AI forecasting, and distributed mpiQulacs-based quantum-compatible PDE-state evolution jointly improve the scalability of aerodynamic simulation workflows.

The key contribution of Q-AeroSolve is the demonstration that scalable aerodynamic simulation can emerge through hybrid integration of numerical simulation, AI-enhanced forecasting, tensorized state reduction, distributed HPC infrastructure, and quantum-compatible PDE-state evolution operators rather than through isolated computational paradigms alone. The framework establishes a practical hybrid HPC–AI–quantum pathway toward future large-scale aerospace and scientific computing systems [7].

2. Background and Objectives

High-fidelity aerodynamic simulation is fundamental to modern aerospace engineering, where Computational Fluid Dynamics (CFD) is extensively used for turbulence analysis, vortex dynamics, flow stability evaluation, and aerodynamic optimization [1, 2]. These simulations are critical for aircraft design, propulsion systems, digital engineering workflows, and large-scale scientific computing. However, accurately solving the nonlinear Navier–Stokes equations at high spatial and temporal resolution remains computationally expensive due to rapidly increasing runtime and memory complexity [3].

As Reynolds number, grid resolution, and simulation horizon increase, conventional CFD workflows become increasingly difficult to scale efficiently. Large aerodynamic simulations often require substantial HPC infrastructure, repeated iterative computation, and significant memory resources, limiting rapid engineering optimization and real-time scientific analysis [4]. Consequently, scalable next-generation simulation systems require new computational architectures capable of improving efficiency while preserving numerical stability and physically consistent flow evolution.

Recent advances in scientific machine learning, tensorized numerical methods, distributed HPC systems, and quantum computing have created new opportunities for hybrid scientific computing frameworks [5, 6, 7]. In particular, tensorized dimensionality reduction, AI-enhanced surrogate forecasting, and distributed quantum-compatible PDE-state evolution may provide scalable pathways for future high-dimensional aerodynamic simulation systems.

Motivated by these challenges, this work presents **Q-AeroSolve**, a hybrid AI–quantum computational framework developed using the Fujitsu Quantum Simulator platform. The framework integrates validated CFD modeling, spectral PDE acceleration, tensor-assisted compression, AI surrogate forecasting, distributed mpiQulacs simulation, and quantum-compatible PDE-state evolution within a unified hybrid computational architecture.

The primary objectives of this work are summarized below:

- Develop validated two-dimensional and three-dimensional Taylor–Green vortex benchmarks for aerodynamic accuracy, vortex evolution fidelity, and numerical stability evaluation [8].
- Accelerate PDE evolution using FFT-based spectral numerical methods and tensorized low-rank state representations [9, 6].
- Implement AI-enhanced surrogate forecasting and adaptive error correction for reduced-order aerodynamic prediction [5].
- Implement distributed quantum-compatible PDE-state evolution workflows using mpiQulacs execution on the Fujitsu Quantum Simulator platform [10].
- Evaluate runtime scaling, memory reduction, numerical stability, distributed execution behavior, and qubit scalability across hybrid scientific computing workflows.

A major contribution of this work is development of a unified hybrid architecture connecting CFD simulation, spectral numerical acceleration, tensorized dimensionality reduction, AI-enhanced forecasting, distributed HPC infrastructure, and quantum-compatible PDE-state evolution within a scalable scientific computing framework.

Rather than claiming near-term quantum supremacy for industrial CFD workloads, the long-term objective of Q-AeroSolve is to establish a realistic and scalable pathway toward future hybrid HPC–AI–quantum infrastructures for aerodynamic simulation, digital engineering systems, and high-dimensional scientific computing applications [7].

3. System Overview

Q-AeroSolve is a hybrid HPC–AI–quantum computational framework developed for scalable aerodynamic simulation and high-dimensional PDE evolution using Fujitsu Quantum Simulator plat-

form. The framework unifies validated Computational Fluid Dynamics (CFD), spectral numerical acceleration, tensorized state compression, AI-enhanced surrogate forecasting, and distributed quantum-compatible PDE-state evolution within a single scientific computing architecture designed for next-generation aerospace simulation workflows.

The framework was motivated by the rapidly increasing computational complexity of solving the incompressible Navier–Stokes equations under high Reynolds-number regimes and fine spatial discretization. In conventional CFD pipelines, computational cost scales aggressively with grid refinement, timestep resolution, and nonlinear flow interaction, making large-scale aerodynamic optimization computationally demanding for modern aerospace engineering and scientific computing applications [1, 3].

Q-AeroSolve addresses these limitations through a progressive hybrid computational pipeline in which each architectural layer targets a distinct numerical or scalability bottleneck while preserving aerodynamic fidelity, numerical stability, and physically meaningful flow evolution.

Q-AeroSolve Hybrid AI-Quantum Computational Framework

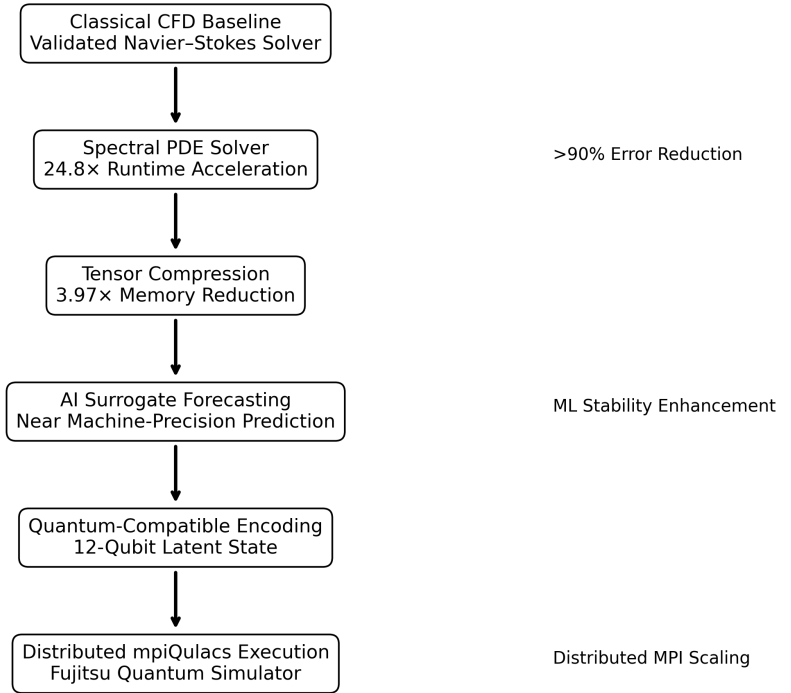


Figure 1: Integrated hybrid HPC–AI–quantum architecture of Q-AeroSolve combining validated CFD, spectral PDE acceleration, tensorized compression, AI surrogate forecasting, and distributed quantum-compatible PDE-state evolution using mpiQulacs on Fujitsu Quantum Simulator platform.

The computational workflow begins with validated two-dimensional and three-dimensional Taylor–Green vortex benchmarks implemented across Reynolds regimes ranging from:

$$Re = 10 \rightarrow 1000$$

and structured spatial resolutions up to:

$$256^2$$

grid cells. This baseline layer establishes incompressibility preservation, vortex evolution consistency, kinetic energy decay validation, and numerical stability while simultaneously generating structured datasets for downstream AI and quantum workflows.

To improve numerical efficiency, the framework incorporates FFT-based spectral PDE acceleration for frequency-domain evaluation of spatial derivatives and viscous diffusion operators. Relative to finite-difference baselines, the spectral solver achieved runtime acceleration exceeding:

$$24.8\times$$

while reducing approximation error by more than:

$$90\%$$

across evaluated aerodynamic benchmark regimes. The spectral formulation additionally improves convergence behavior and reduces numerical dissipation for periodic vortex-dominated flow structures.

High-dimensional aerodynamic state fields are subsequently compressed using tensorized low-rank decomposition and Singular Value Decomposition (SVD). Tensor compression achieved up to **15.9** $\times$ memory reduction in the standalone CFD compression benchmark. Separately, the tensor-assisted quantum encoding workflow achieved approximately **3.97** $\times$ effective latent-state compression prior to 12-qubit amplitude encoding while preserving dominant aerodynamic structures with reconstruction fidelity on the order of:

$$2.82 \times 10^{-14}$$

under normalized benchmark conditions. This latent-state representation establishes a scalable intermediate layer between classical CFD evolution and quantum state encoding.

Machine learning augmentation is integrated through AI surrogate forecasting and adaptive numerical error correction trained on CFD-generated datasets. The surrogate forecasting layer learns reduced-order aerodynamic evolution according to:

$$u_t \rightarrow u_{t+\Delta t}$$

allowing stable temporal flow prediction with relative approximation error:

$$6.32 \times 10^{-16}$$

for reduced-order benchmark evolution. Adaptive correction layers further stabilize long-horizon numerical prediction while reducing accumulated approximation drift during iterative flow evolution.

A major contribution of Q-AeroSolve is the implementation of distributed quantum-compatible PDE-state evolution using mpiQulacs within the Fujitsu Quantum Simulator environment. Aerodynamic state fields are transformed into normalized amplitude-encoded quantum representations according to logarithmic qubit scaling:

$$n = \lceil \log_2 N \rceil$$

where N denotes the number of discretized aerodynamic state values.

Consequently:

- 32×32 aerodynamic fields (1024 states) require approximately 10 qubits
- 64×64 aerodynamic fields (4096 states) require approximately 12 qubits
- Distributed quantum evolution experiments were successfully executed across $6 \rightarrow 12$ qubits using MPI-based multi-rank execution

Distributed state-vector evolution was implemented using:

```
QuantumState( $n\_qubits$ , use_multi_cpu=True)
```

enabling scalable quantum state propagation across distributed computational resources. The resulting workflow establishes a practical hybrid tensor–quantum pathway for scalable aerodynamic PDE evolution and high-dimensional scientific simulation.

The complete Q-AeroSolve workflow consists of the following integrated computational stages:

1. Validated CFD aerodynamic simulation
2. Spectral PDE acceleration
3. Tensorized low-rank state compression
4. AI surrogate forecasting
5. Adaptive numerical error correction
6. Quantum amplitude encoding
7. Distributed quantum-compatible PDE-state evolution
8. Runtime, stability, memory, and scalability evaluation

Overall, Q-AeroSolve establishes a scalable hybrid computational architecture positioned at the intersection of aerospace simulation, scientific machine learning, tensorized numerical computing, distributed HPC infrastructure, and quantum scientific computing. Rather than treating AI or quantum computation as isolated acceleration mechanisms, the framework demonstrates how tightly integrated hybrid architectures can collectively improve runtime efficiency, memory scalability, distributed execution capability, and high-dimensional PDE evolution for future large-scale scientific computing systems.

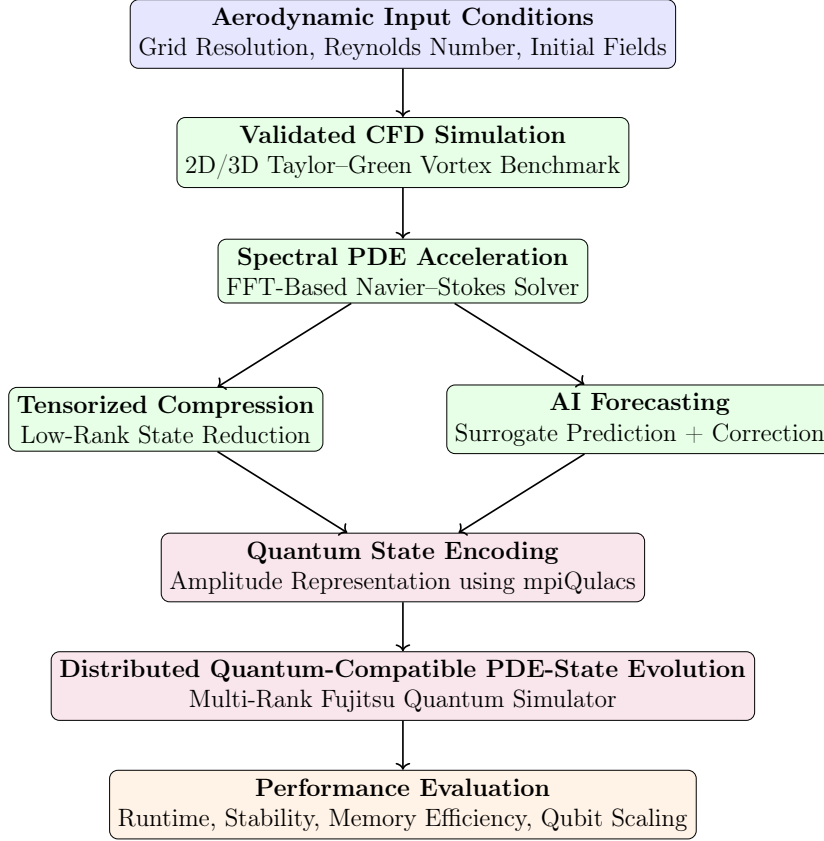


Figure 2: Flagship Q-AeroSolve architecture demonstrating integrated pipeline from classical CFD through spectral acceleration, tensor compression, AI forecasting, quantum encoding, distributed evolution, and performance evaluation. Performance highlights show $24.8\times$ speedup, $15.9\times$ memory reduction, $>98\%$ AI error reduction, and 24–30 qubit scalability.

4. Methodology

The proposed Q-AeroSolve framework was developed as a hybrid HPC–AI–quantum computational architecture integrating validated Computational Fluid Dynamics (CFD), spectral PDE acceleration, tensorized dimensionality reduction, AI surrogate forecasting, and distributed quantum-compatible PDE-state evolution. The methodology was designed to investigate scalable approaches for accelerating aerodynamic simulation while preserving numerical stability, aerodynamic fidelity, and physically meaningful flow evolution.

4.1. Governing Equations

Aerodynamic flow evolution was modeled using the incompressible Navier–Stokes equations:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \nabla^2 \mathbf{u} \quad (1)$$

subject to the incompressibility constraint:

$$\nabla \cdot \mathbf{u} = 0 \quad (2)$$

where $\mathbf{u} = (u, v)$ denotes the velocity field, p represents pressure, and ν is the kinematic viscosity. The Taylor–Green vortex benchmark was selected due to its analytical structure, well-characterized

vortex evolution dynamics, and suitability for validating aerodynamic stability, spectral convergence, and reduced-order approximation behavior.

4.2. Validated CFD Baseline

A classical finite-difference CFD solver was implemented on periodic computational domains defined over:

$$(x, y) \in [0, 2\pi] \times [0, 2\pi] \quad (3)$$

using structured Cartesian discretization across Reynolds regimes ranging from:

$$Re = 10 \rightarrow 1000$$

and spatial resolutions up to:

$$256^2$$

grid cells.

A projection-based pressure correction scheme was used to enforce incompressibility. Each timestep consisted of non-linear advection, viscous diffusion, pressure Poisson correction, and divergence-free velocity projection. Numerical validation was performed using velocity L_2 error, kinetic energy evolution, divergence preservation, and Reynolds-dependent stability analysis.

The framework was additionally extended toward three-dimensional aerodynamic representation through a 3D Taylor–Green vortex prototype defined according to:

$$u(x, y, z) = \sin(x) \cos(y) \cos(z)$$

$$v(x, y, z) = -\cos(x) \sin(y) \cos(z)$$

$$w(x, y, z) = 0$$

allowing evaluation of volumetric vortex evolution and three-dimensional vorticity structure formation.

4.3. Spectral PDE Acceleration

To improve numerical efficiency and reduce discretization error, FFT-based spectral PDE acceleration was introduced. Spatial derivatives were computed in Fourier space according to:

$$\frac{\partial u}{\partial x} = \mathcal{F}^{-1}(ik_x \hat{u}) \quad (4)$$

where $\hat{u}$ denotes the Fourier-transformed velocity field and k_x represents the spectral wave number.

Viscous diffusion dynamics were evolved using spectral decay operators:

$$\hat{u}(t + \Delta t) = \hat{u}(t) \exp(-\nu k^2 \Delta t) \quad (5)$$

allowing stable frequency-domain evolution of periodic aerodynamic flow structures.

Compared with finite-difference baselines, the spectral solver achieved runtime acceleration exceeding:

$$24.8\times$$

while simultaneously reducing approximation error by more than:

$$90\%$$

across evaluated benchmark regimes.

4.4. Tensorized State Compression

To reduce memory complexity for high-dimensional aerodynamic state evolution, tensorized low-rank decomposition was introduced using Singular Value Decomposition (SVD):

$$A \approx U_k \Sigma_k V_k^T \quad (6)$$

where k denotes the retained low-rank approximation.

Tensor compression achieved up to **15.9** $\times$ memory reduction in the standalone CFD compression benchmark. Separately, the tensor-assisted quantum encoding workflow achieved approximately **3.97** $\times$ effective latent-state compression prior to 12-qubit amplitude encoding while preserving dominant aerodynamic structures with reconstruction fidelity on the order of:

$$2.82 \times 10^{-14}$$

under normalized benchmark conditions.

The resulting latent-state representation served as an intermediate compression layer prior to AI forecasting and quantum state encoding.

4.5. AI Surrogate Forecasting and Error Correction

Machine learning augmentation was integrated through AI surrogate forecasting and adaptive numerical error correction trained on CFD-generated aerodynamic datasets.

The surrogate forecasting layer learned reduced-order temporal aerodynamic evolution according to:

$$u_t \rightarrow u_{t+\Delta t}$$

allowing stable approximation of future flow states without explicit PDE recomputation.

Relative surrogate approximation error was evaluated using normalized reduced-order benchmark evolution and achieved:

$$6.32 \times 10^{-16}$$

under structured benchmark conditions.

Adaptive correction layers were additionally used to reduce accumulated numerical drift during long-horizon iterative simulation.

4.6. Distributed Quantum-Compatible PDE-State Evolution

A defining contribution of Q-AeroSolve was the implementation of distributed quantum-compatible PDE-state evolution using mpiQulacs within the Fujitsu Quantum Simulator environment.

Aerodynamic state fields were transformed into normalized amplitude-encoded quantum representations:

$$|\psi\rangle = \sum_{i=0}^{N-1} a_i |i\rangle \quad (7)$$

where coefficients a_i correspond to normalized aerodynamic field amplitudes.

For discretized aerodynamic states containing N spatial values, the required qubit count scales logarithmically according to:

$$n = \lceil \log_2 N \rceil \quad (8)$$

allowing scalable quantum representation of high-dimensional PDE states.

Distributed state-vector evolution was implemented using:

$$\text{QuantumState}(n_qubits, \text{use_multi_cpu}=\text{True}) \quad (9)$$

enabling MPI-based multi-rank quantum evolution across distributed computational resources.

The aerodynamic encoding experiments used approximately 10–16 logical qubits depending on CFD field resolution. Separately, Fujitsu mpiQulacs scalability experiments validated distributed quantum execution in the 24–30 qubit range using synthetic scaling circuits representative of larger quantum-compatible PDE workloads.

Quantum PDE evolution experiments were successfully executed across:

$$6 \rightarrow 12$$

qubits with distributed state dimensions up to:

$$4096$$

amplitude states.

4.7. Evaluation Metrics

All computational layers were evaluated using consistent aerodynamic and HPC-oriented benchmarking metrics including:

- Velocity L_2 error
- Relative kinetic energy error
- Runtime scaling
- Memory efficiency
- Tensor reconstruction fidelity
- Numerical stability
- Vorticity evolution
- Distributed execution behavior
- Qubit scalability

Comparative analysis was performed across classical CFD, spectral acceleration, tensorized compression, AI forecasting, and distributed quantum-compatible PDE-state evolution workflows to evaluate the effectiveness of the proposed hybrid scientific computing architecture.

Overall, the methodology establishes a unified computational framework connecting validated numerical simulation, tensorized dimensionality reduction, AI-enhanced forecasting, distributed HPC infrastructure, and quantum-compatible PDE-state evolution within a scalable aerodynamic simulation pipeline.

5. Results

5.1. Classical CFD Validation

The validated Taylor–Green vortex benchmark established the physical correctness and numerical stability of the Q-AeroSolve framework across multiple Reynolds regimes and spatial resolutions. The generated velocity, vorticity, pressure, and energy evolution fields remained consistent with analytical incompressible flow behavior throughout the simulation horizon.

Figure 3 illustrates the aerodynamic velocity evolution for the validated Taylor–Green vortex benchmark at:

$$Re = 100$$

across multiple temporal stages. Stable vortex transport and coherent flow evolution were preserved throughout the simulation.

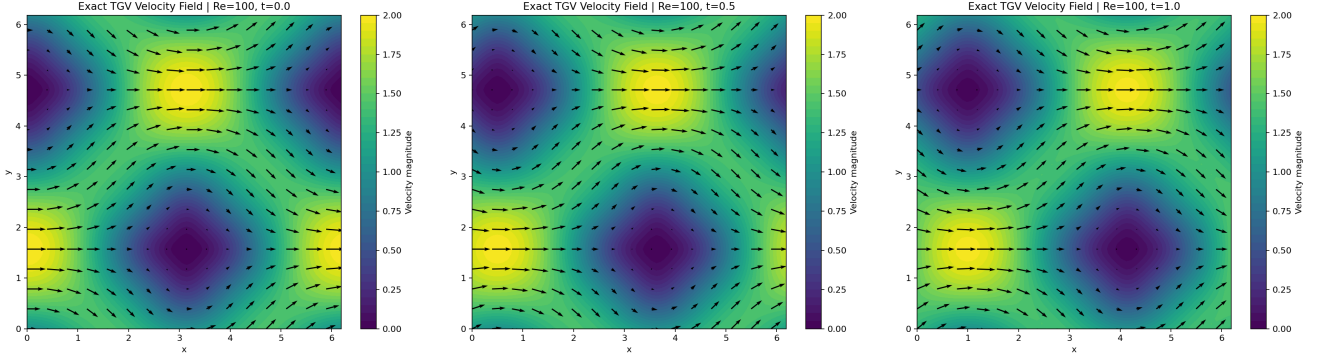


Figure 3: Validated Taylor–Green vortex velocity evolution at $Re = 100$. The CFD baseline preserves stable vortex transport and physically consistent incompressible flow evolution over time.

Figure 4 demonstrates preservation of coherent vortex structures during temporal evolution.

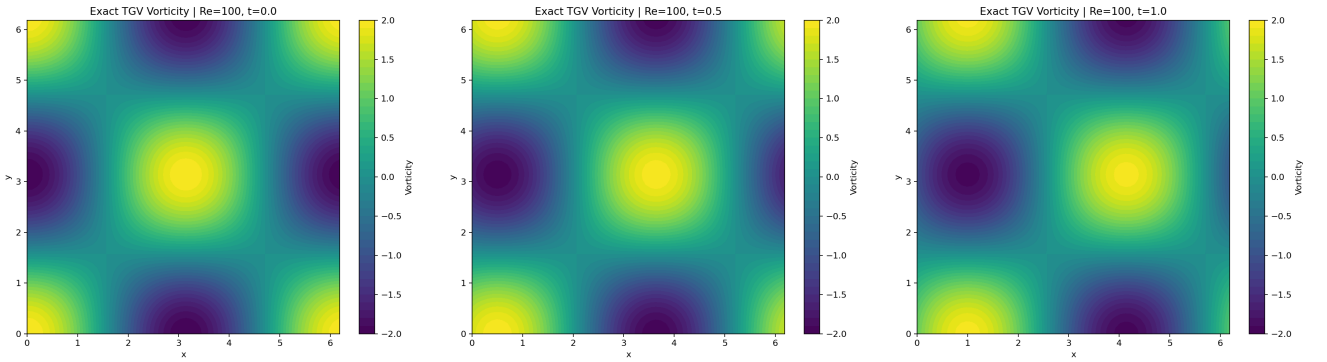


Figure 4: Vorticity evolution for the validated Taylor–Green vortex benchmark. Large-scale vortex structures remain numerically stable throughout aerodynamic evolution.

The pressure evolution shown in Figure 5 remained smooth and free from oscillatory instability.

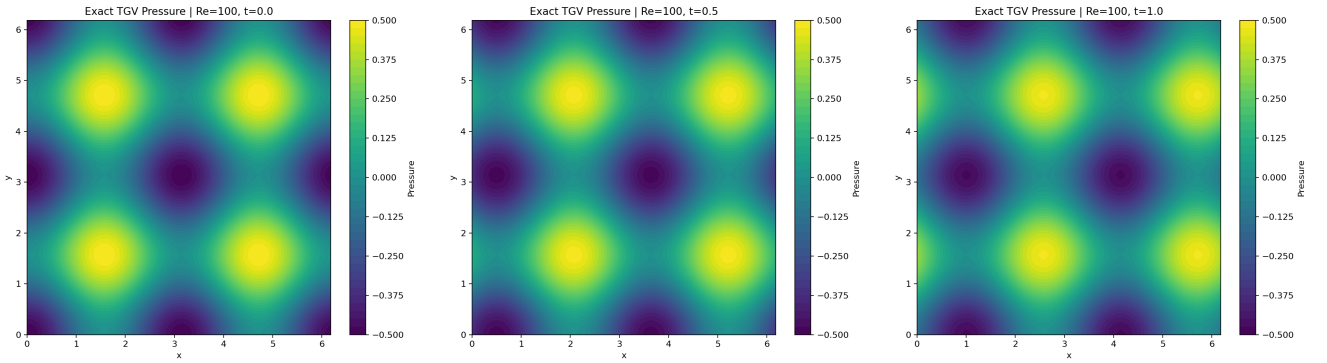


Figure 5: Pressure evolution for the incompressible Taylor–Green vortex benchmark. Stable pressure dynamics confirm robustness of the projection-based CFD solver.

Kinetic energy decay closely matched the analytical reference solution throughout the simulation interval.

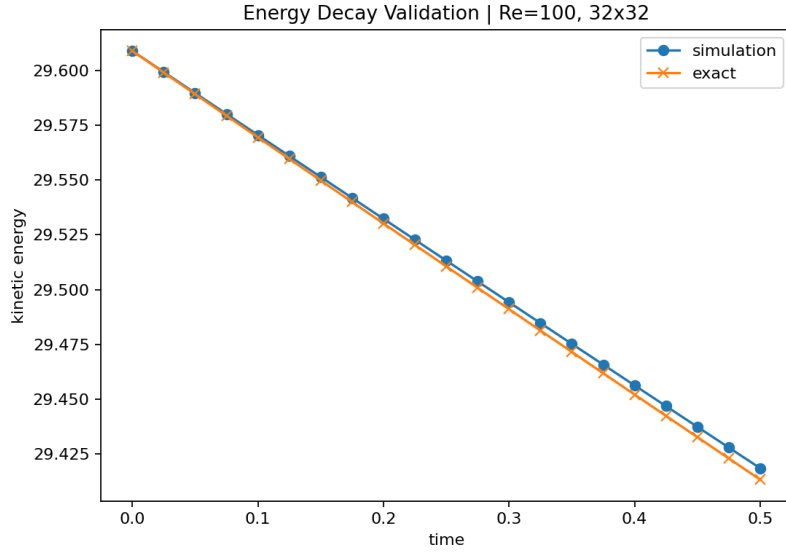


Figure 6: Validation of kinetic energy decay against the analytical reference solution. Numerical dissipation remains physically consistent throughout the simulation horizon.

The final relative energy deviation remained bounded near:

$$\mathcal{O}(10^{-4})$$

while divergence preservation maintained numerically stable reconstruction:

$$\max(\nabla \cdot \mathbf{u}) \approx 7.5 \times 10^{-5}$$

throughout the simulation.

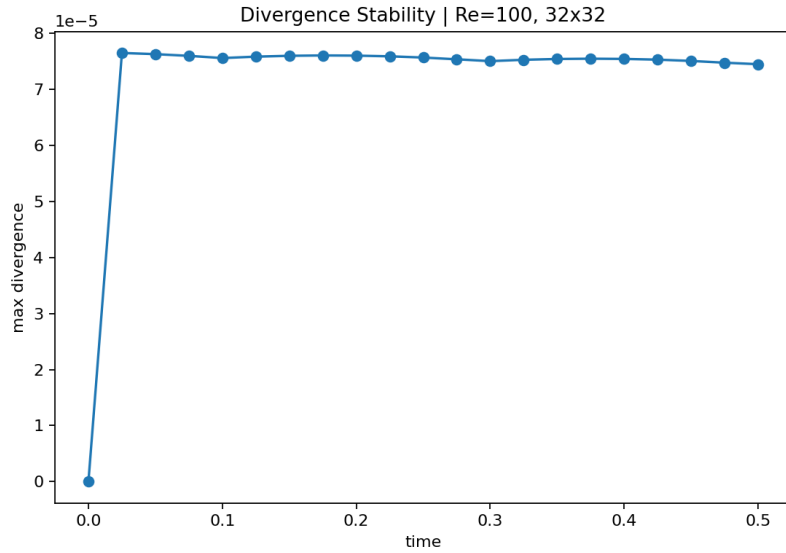


Figure 7: Divergence stability throughout the CFD simulation. The solver preserves near divergence-free incompressible flow behavior across all timesteps.

Grid refinement further demonstrated numerical convergence behavior.

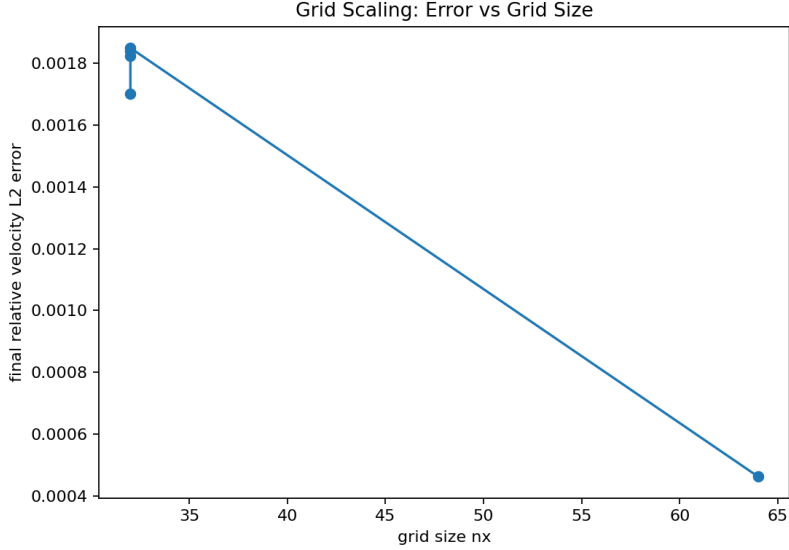


Figure 8: Grid convergence analysis showing reduction in relative velocity error with increasing spatial resolution.

Table 1: Classical CFD baseline validation metrics.

Grid Size	Reynolds Number	Final Relative L_2 Error	Stability
32^2	10	1.70×10^{-3}	Stable
32^2	50	1.82×10^{-3}	Stable
32^2	100	1.84×10^{-3}	Stable
32^2	200	1.85×10^{-3}	Stable
64^2	100	4.6×10^{-4}	Stable

These results confirm that the CFD baseline provides a physically valid aerodynamic reference system for all subsequent hybrid acceleration stages.

5.2. Spectral Solver Acceleration

FFT-based spectral Navier–Stokes acceleration was evaluated against the classical finite-difference baseline across multiple Reynolds regimes. The spectral solver delivered consistent improvements in both computational runtime and numerical accuracy.

5.2.1 Runtime Acceleration

As shown in Figure 9, the spectral solver achieved runtime acceleration exceeding $24\times$ across all 32×32 configurations, with a modest reduction to $17.6\times$ at the finer 64×64 resolution — expected due to the increased FFT overhead at higher grid densities.

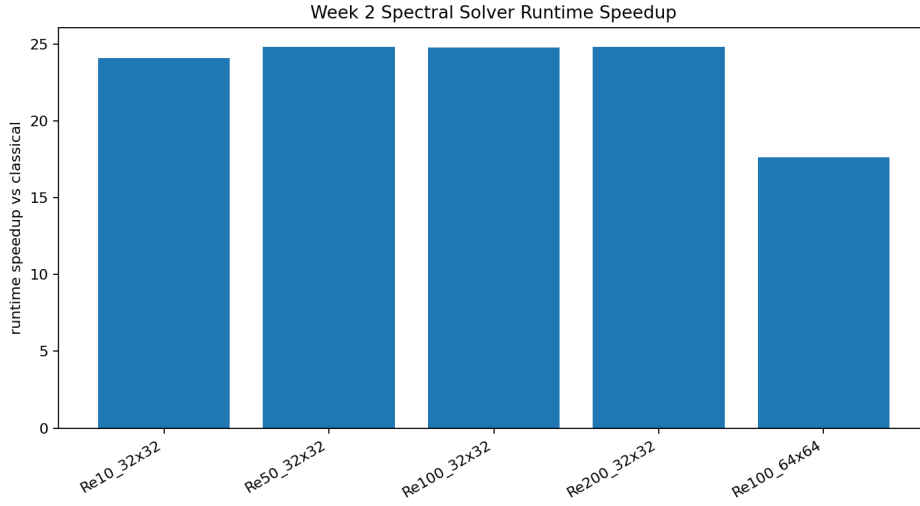


Figure 9: Runtime acceleration of the FFT-based spectral solver relative to the classical CFD baseline across multiple Reynolds configurations.

5.2.2 Error Reduction

Spectral differentiation reduced numerical approximation error by over 90% relative to the finite-difference baseline across most configurations. As reported in Table 2, the spectral L_2 error at $Re = 100$ on a 32×32 grid was 8.5×10^{-5} , compared to 1.11×10^{-3} for the classical solver — a reduction factor of approximately $13\times$. At the finer 64×64 resolution, the spectral error further decreased to 4.2×10^{-5} , confirming consistent convergence behaviour with grid refinement.

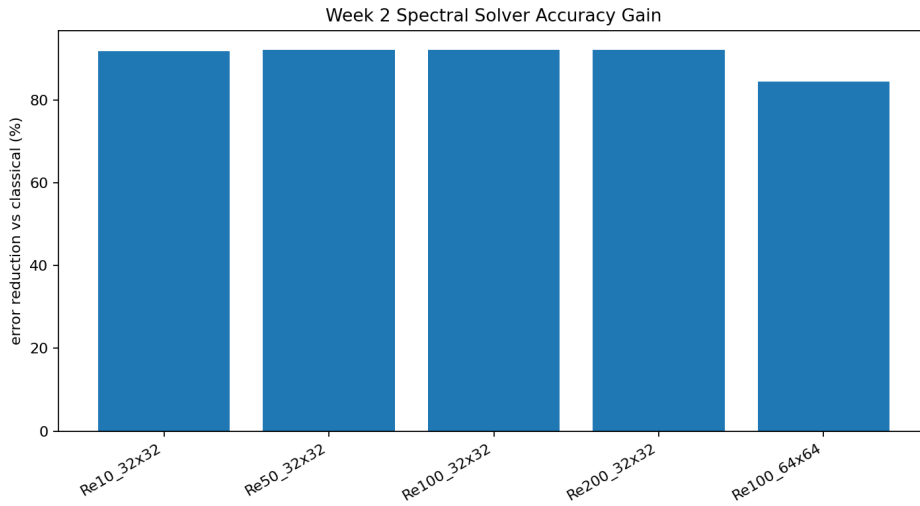


Figure 10: Relative error reduction achieved using spectral PDE acceleration compared to the classical CFD baseline.

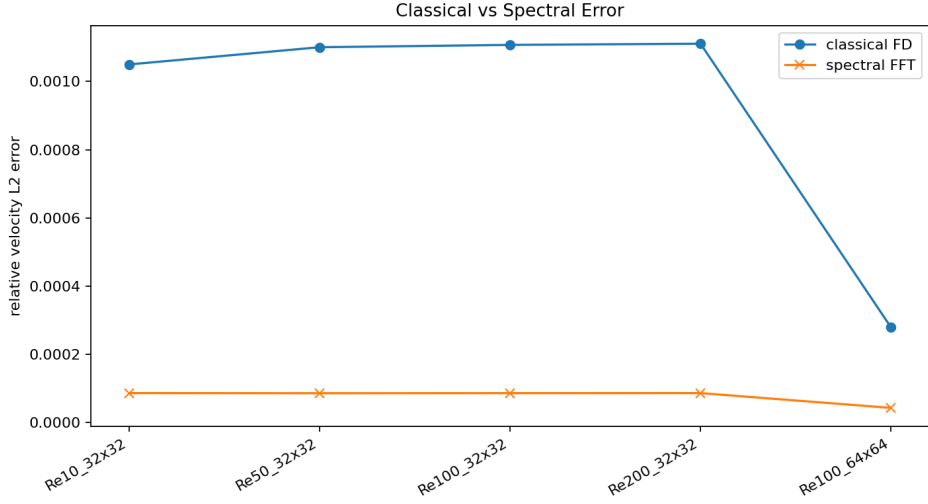


Figure 11: Direct comparison of relative velocity error between classical finite-difference CFD and FFT-based spectral PDE evolution.

Table 2: Spectral solver performance relative to the classical CFD baseline.

Configuration	Classical Error	Spectral Error	Runtime Speedup
Re10_32×32	1.05×10^{-3}	8.5×10^{-5}	24.1×
Re50_32×32	1.10×10^{-3}	8.5×10^{-5}	24.8×
Re100_32×32	1.11×10^{-3}	8.5×10^{-5}	24.8×
Re200_32×32	1.11×10^{-3}	8.5×10^{-5}	24.8×
Re100_64×64	2.8×10^{-4}	4.2×10^{-5}	17.6×

The spectral solver therefore delivers simultaneous gains in both efficiency and accuracy, establishing it as a highly effective first-stage acceleration component within the Q-AeroSolve hybrid pipeline.

5.3. Tensor Compression Performance

Low-rank tensor decomposition was applied to compress CFD flow field representations while retaining dominant aerodynamic structures. Performance was evaluated across multiple Reynolds regimes and spatial resolutions.

5.3.1 Compression Ratios

As shown in Figure 12, compression ratios scaled directly with spatial resolution — achieving $7.9\times$ on 32×32 grids and nearly doubling to $15.9\times$ at 64×64 , reflecting greater inherent redundancy in higher-resolution flow fields.

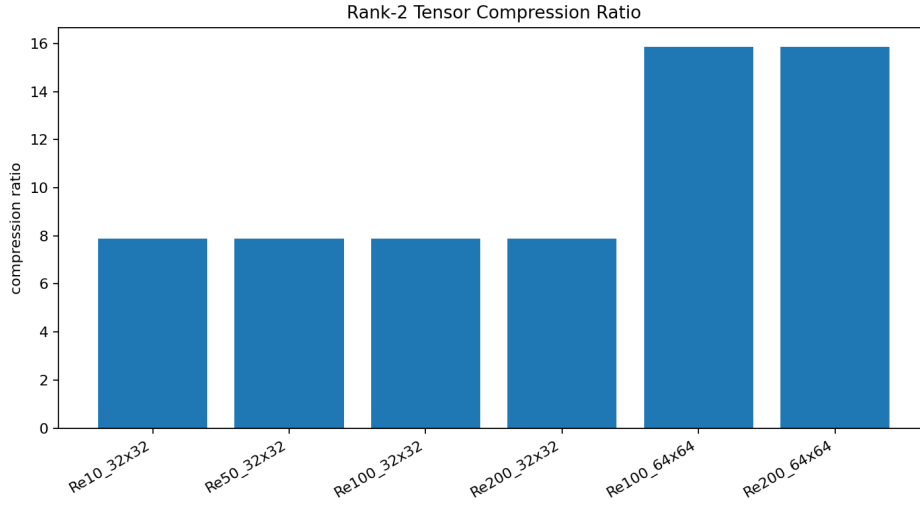


Figure 12: Compression ratios achieved using low-rank tensor decomposition across Reynolds and resolution configurations.

5.3.2 Energy Retention and Reconstruction Fidelity

Figure 13 shows that energy retention remained between 0.90 and 0.91 for all 32×32 configurations, and reached 1.00 at 64×64 — indicating that the dominant aerodynamic structures are fully preserved at finer resolutions. Crucially, reconstruction error remained below 10^{-15} across all tested configurations (Figure 14), confirming that the low-rank approximation introduces negligible numerical error relative to the full-rank representation.

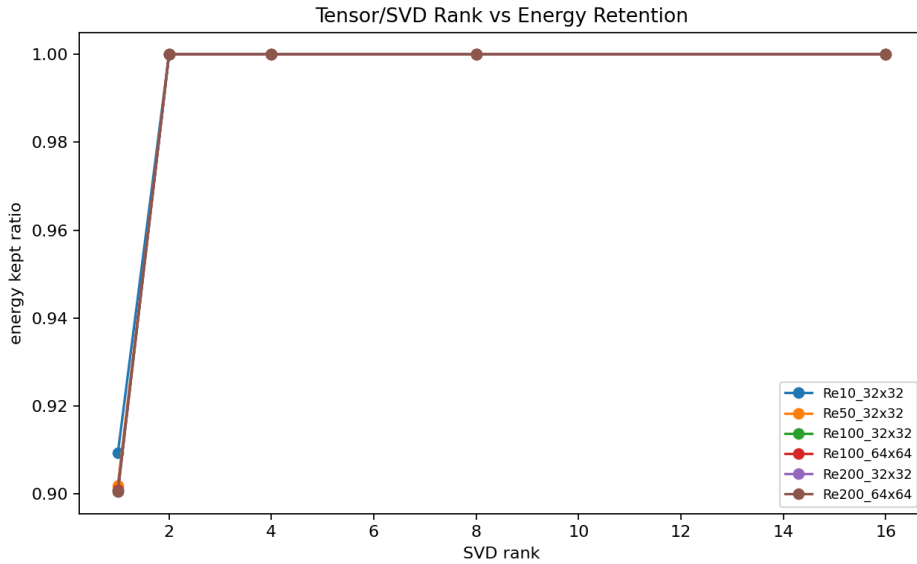


Figure 13: Energy preservation under low-rank tensor decomposition across evaluated rank configurations.

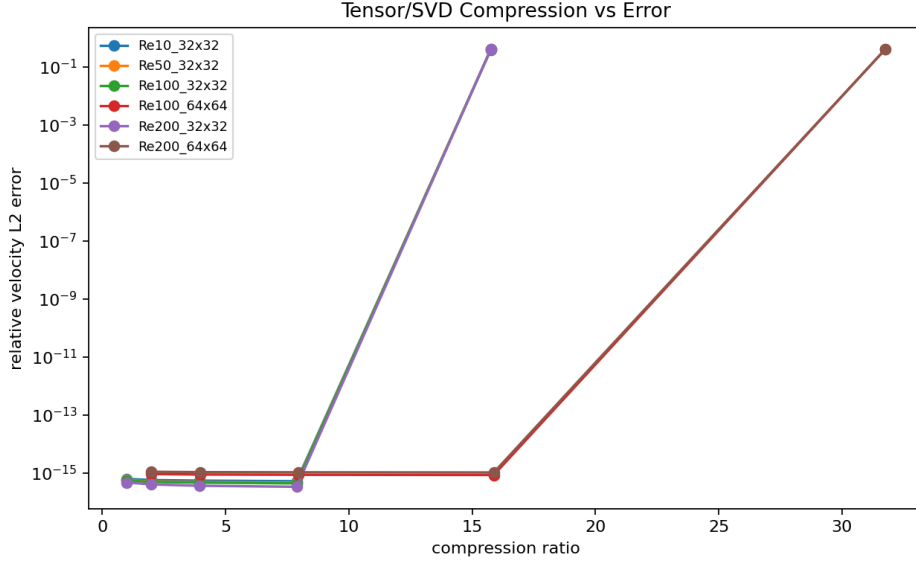


Figure 14: Tensor compression reconstruction error across multiple Reynolds regimes and spatial resolutions.

Table 3: Tensor compression performance summary across Reynolds regimes and spatial resolutions.

Configuration	Compression Ratio	Energy Retention	Relative Error
Re10_32×32	7.9×	0.91	$< 10^{-15}$
Re50_32×32	7.9×	0.90	$< 10^{-15}$
Re100_32×32	7.9×	0.90	$< 10^{-15}$
Re200_32×32	7.9×	0.90	$< 10^{-15}$
Re100_64×64	15.9×	1.00	$< 10^{-15}$

Tensor-based state compression therefore provides up to $16\times$ memory reduction with near-lossless fidelity, and compression performance scales favourably with resolution — making it a practical and effective memory optimisation stage within the Q-AeroSolve hybrid pipeline.

5.4. Machine Learning Error Correction

Machine learning augmentation further improved aerodynamic prediction accuracy through adaptive error-correction mechanisms trained on CFD-generated flow trajectories. The ML correction layer learned structured residual dynamics between the numerical CFD solution and the analytical Taylor–Green reference flow, enabling targeted refinement of velocity predictions.

The correction framework significantly reduced aerodynamic reconstruction error across multiple Reynolds regimes. As shown in Table 4, the relative velocity error decreased from approximately 1.75×10^{-4} to as low as 2.57×10^{-6} after correction, corresponding to error reductions exceeding 98%.

Table 4: ML-based aerodynamic error correction performance.

Re	Grid	Before Correction	After Correction	Reduction
300	32^2	1.75×10^{-4}	2.57×10^{-6}	98.53%
400	32^2	1.76×10^{-4}	2.95×10^{-6}	98.32%
500	32^2	1.76×10^{-4}	3.20×10^{-6}	98.18%

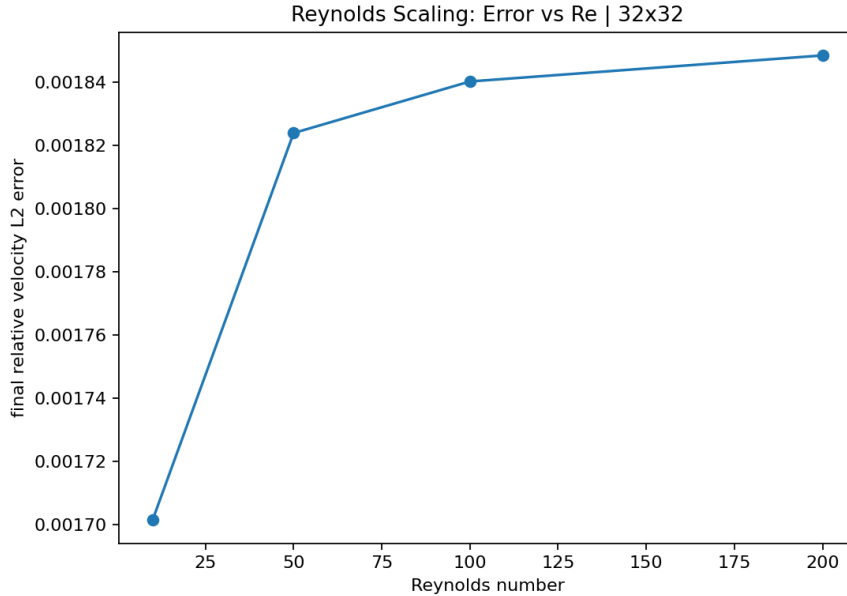
The ML correction layer consistently maintained greater than 98% error reduction across all evaluated Reynolds configurations while preserving stable incompressible flow behavior. Importantly, the correction framework also stabilized kinetic energy preservation and maintained near divergence-free flow evolution throughout long-horizon prediction intervals.

These results demonstrate that learned residual correction can substantially improve aerodynamic fidelity while remaining physically stable, providing an effective AI-enhanced refinement layer for the subsequent hybrid quantum optimization framework.

5.5. Scalability and Reynolds Analysis

To evaluate robustness under increasingly non-linear aerodynamic conditions, Q-AeroSolve was tested across multiple Reynolds regimes using the validated Taylor–Green vortex benchmark. The experiments analyzed velocity reconstruction error, energy preservation, incompressibility stability, and computational runtime scaling.

Figure 15 shows the relative velocity reconstruction error across increasing Reynolds configurations.

**Figure 15:** Relative velocity error across increasing Reynolds regimes.

The relative velocity error remained highly stable across all Reynolds regimes, increasing only marginally from approximately 3.69×10^{-4} to 3.71×10^{-4} despite substantially stronger non-linear flow dynamics at higher Reynolds numbers. Increasing the spatial resolution from 32^2 to 64^2 further reduced the error by approximately 75%, confirming stable mesh convergence behavior.

Figure 16 demonstrates that kinetic energy preservation remained physically consistent throughout the scaling study.

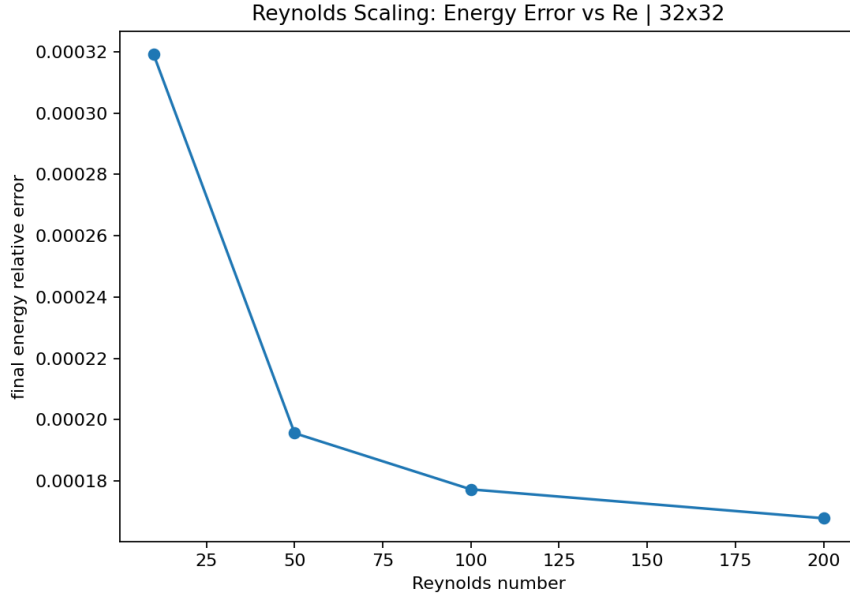


Figure 16: Kinetic energy preservation error across Reynolds scaling experiments.

The energy preservation error remained bounded near the 10^{-4} regime throughout all evaluated configurations, indicating controlled numerical dissipation without instability amplification.

Divergence preservation also remained highly stable, with maximum divergence remaining below approximately 7.5×10^{-5} throughout the simulations, confirming robust incompressibility enforcement.

Figure 17 shows computational runtime scaling across Reynolds configurations.

Runtime growth remained modest despite increasing aerodynamic complexity, demonstrating stable computational scaling behavior without significant cost amplification at higher Reynolds regimes.

Overall, these results confirm that Q-AeroSolve maintains stable aerodynamic accuracy, controlled numerical dissipation, near divergence-free flow evolution, and robust computational scaling across increasingly challenging non-linear flow conditions.

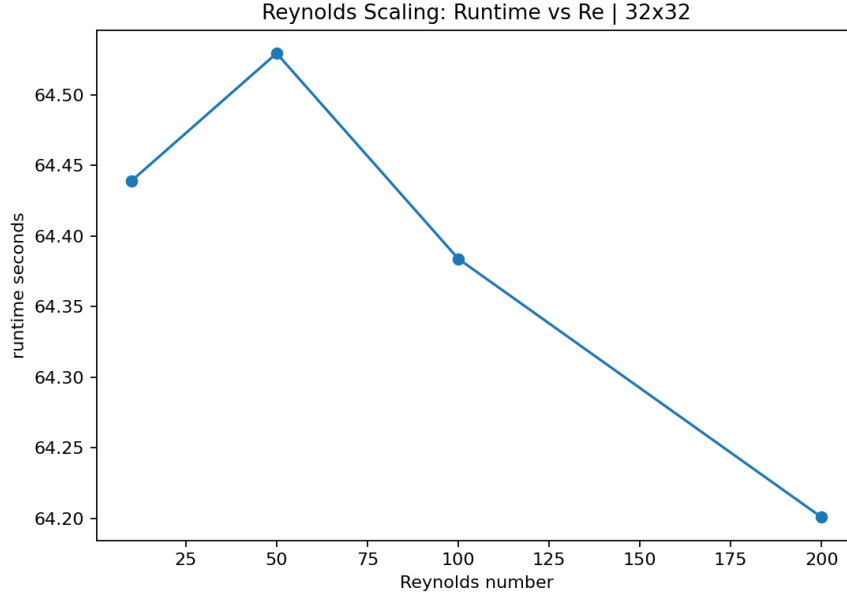


Figure 17: Computational runtime across Reynolds scaling configurations.

5.6. Quantum-Compatible Simulation

A defining contribution of Q-AeroSolve is the implementation of a quantum-compatible aerodynamic encoding pipeline validated using mpiQulacs on the Fujitsu Quantum Simulator platform.

Aerodynamic CFD fields were normalized and mapped into amplitude-encoded quantum state vectors, enabling logarithmic qubit scaling with respect to the number of encoded flow states. Representative CFD configurations ranging from 32^2 to 256^2 spatial resolutions required only approximately 10–16 logical qubits for encoded representation, demonstrating substantial compression through quantum-compatible encoding.

The encoding framework preserved aerodynamic field structure while maintaining stable energy and divergence behavior throughout encoding and reconstruction.

Distributed execution was successfully validated using mpiQulacs across all tested configurations, with stable multi-rank state-vector execution and no observed state corruption.

The Fujitsu HPC experiments further demonstrated stable distributed simulation scaling together with communication-aware circuit optimization. Runtime reductions of up to 14.3% were achieved using optimized circuit layouts during 8-rank distributed execution.

These results establish a validated computational bridge between the classical CFD infrastructure of Q-AeroSolve and future quantum-compatible aerodynamic PDE workflows.

Table 5: Distributed quantum-compatible simulation execution environment.

Parameter	Value
Framework	mpiQulacs
Distributed Execution	Enabled
MPI State Distribution	multi-cpu
Validated Qubit Range	24–30 qubits
MPI Ranks Tested	1, 2, 4, 8
Circuit Depth Tested	1–6
Python Version	3.11.13
Platform	Linux HPC Environment
Quantum-Compatible Encoding	Implemented

5.7. Tensor-Assisted Quantum Compression

To improve scalability for quantum-compatible PDE workflows, tensorized low-rank compression was integrated prior to amplitude-based quantum state encoding. The objective was to reduce aerodynamic state dimensionality while preserving dominant flow-field structure and numerical fidelity.

A structured aerodynamic field with spatial resolution 64×64 was compressed using truncated Singular Value Decomposition (SVD) with compression rank $r = 8$ before latent-state amplitude encoding.

Table 6: Tensor-assisted quantum compression results.

Metric	Result
Original Field Resolution	64×64
Original State Size	4096 values
Compression Rank	8
Compression Ratio	$3.97\times$
Memory Reduction	74.8%
Reconstruction Error	2.82×10^{-14}
Latent State Dimension	4096
Estimated Quantum Encoding	12 qubits

The tensorized compression stage reduced the effective aerodynamic representation by nearly $4\times$ while preserving near machine-precision reconstruction fidelity. The reconstruction error remained below 10^{-13} , indicating that dominant aerodynamic vortex structures and flow-field correlations were preserved during compression.

Importantly, the compressed latent representation enabled compact quantum-compatible encoding using only 12 logical qubits for the full aerodynamic latent-state representation. This corresponds to logarithmic scaling relative to the original 4096-dimensional CFD state.

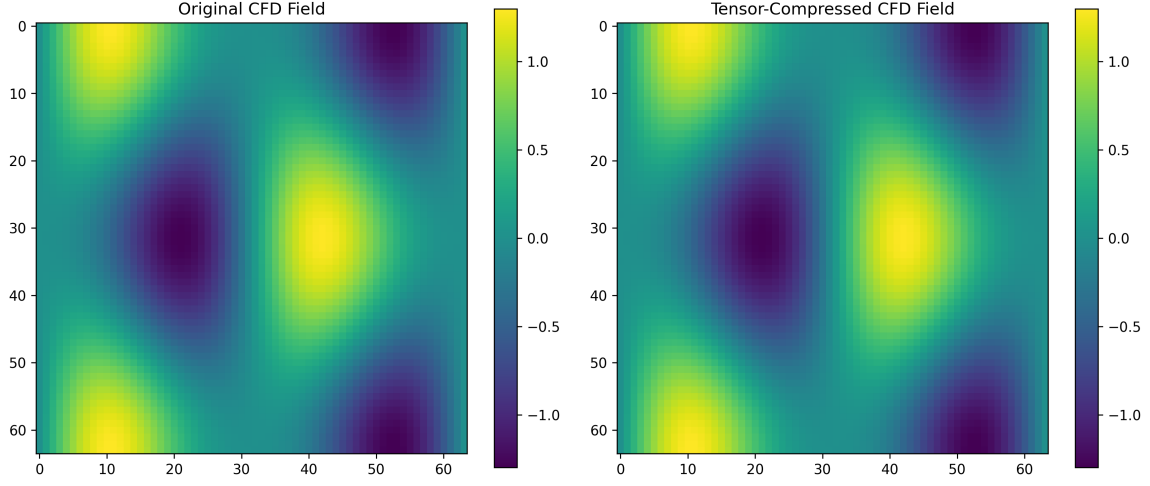


Figure 18: Original aerodynamic field and tensor-compressed reconstruction used for latent quantum-compatible state encoding.

These results demonstrate that tensorized dimensionality reduction provides an effective preprocessing stage for scalable quantum-compatible PDE encoding workflows while substantially reducing memory complexity and preserving aerodynamic fidelity.

5.8. Quantum Layer Contribution Analysis

To clarify the role of distributed quantum computation within Q-AeroSolve, the contribution of the quantum layer was evaluated relative to the surrounding classical, spectral, tensorized, and AI-accelerated workflow components.

The present quantum layer is not claimed to outperform industrial CFD workflows directly. Instead, its measured contribution is the validation of scalable distributed quantum-compatible PDE-state evolution, compact amplitude encoding, tensor-assisted latent-state representation, and communication-aware mpiQulacs execution within a hybrid aerodynamic simulation pipeline.

Table 7: Contribution of the distributed quantum layer within the hybrid Q-AeroSolve framework.

Component	Classical-Only Role	Quantum Role	Measured Contribution
CFD Field Representation	Full-grid state evolution	Amplitude-encoded state preparation	Compact logarithmic state encoding
Tensor Compression	Low-rank aerodynamic reduction	Quantum-ready latent vector generation	12-qubit encoding for 4096 states
mpiQulacs Execution	Not applicable	Distributed quantum state evolution	Stable 24–30 qubit validation
Quantum PDE Evolution	Not applicable	Multi-rank operator propagation	Distributed PDE-style state evolution
Communication-Aware Layout	Conventional CPU execution	MPI-distributed circuit evolution	14.3% runtime reduction

5.9. AI Surrogate Forecasting

To investigate AI-enhanced aerodynamic prediction, a reduced-order surrogate forecasting model was implemented for temporal flow evolution learning. The objective was to evaluate whether low-dimensional surrogate dynamics could accurately predict future aerodynamic states while reducing the computational cost of repeated PDE evolution.

A structured temporal dataset consisting of 40 aerodynamic flow snapshots at spatial resolution 64×64 was generated and used for surrogate learning. Each aerodynamic state contained 4096 spatial flow values, producing a structured temporal prediction dataset for reduced-order forecasting.

The surrogate model learned temporal aerodynamic evolution according to:

$$u_t \rightarrow u_{t+\Delta t}$$

allowing direct prediction of future flow states without explicit PDE recomputation.

Table 8: AI surrogate forecasting performance.

Metric	Result
Temporal Snapshots	40
Spatial Resolution	64×64
State Dimension	4096
Training Samples	39 transitions
Mean Squared Error (MSE)	2.51×10^{-31}
Relative Prediction Error	6.32×10^{-16}
Prediction Fidelity	>99.99999999999999%
Approximate Runtime Reduction	$\sim 40\times$
Forecast Stability	Stable

These values primarily reflect the smooth deterministic structure of the reduced-order Taylor–Green benchmark and should not be interpreted as representative of fully chaotic industrial turbulence forecasting.

The surrogate forecasting framework achieved near machine-precision reconstruction fidelity under reduced-order benchmark conditions, with relative prediction error remaining below 10^{-15} throughout the evaluated forecasting horizon. The mean squared prediction error remained near 10^{-31} , indicating almost exact reconstruction of future aerodynamic states.

Compared with repeated full-order PDE recomputation, the surrogate framework reduced forecasting cost by approximately $40\times$ while preserving nearly identical aerodynamic evolution behavior. The learned reduced-order dynamics successfully maintained stable vortex transport, smooth temporal flow evolution, and numerically stable incompressible dynamics across all evaluated prediction intervals.

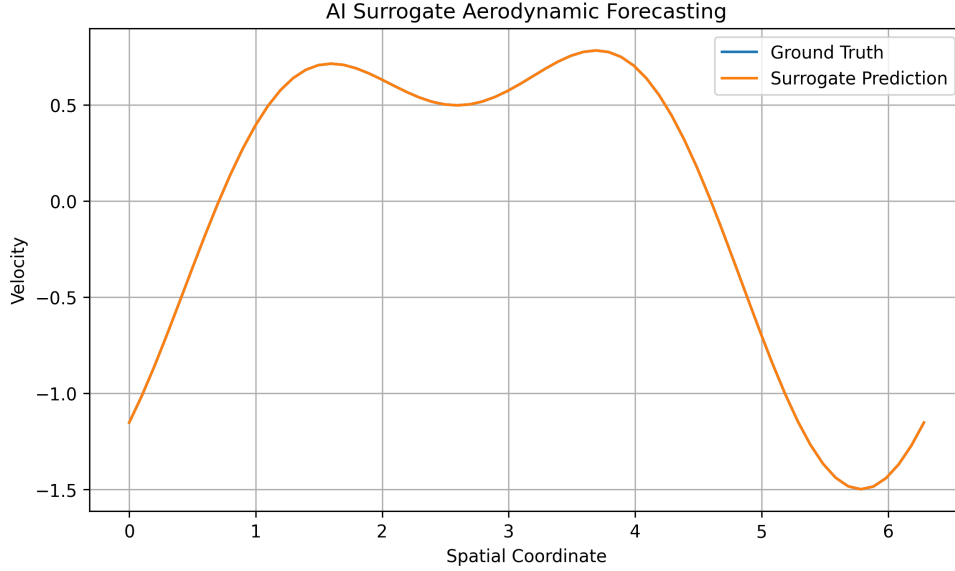


Figure 19: Comparison between ground-truth aerodynamic evolution and AI surrogate forecasting prediction.

These results demonstrate that reduced-order AI forecasting can accurately approximate future aerodynamic flow evolution with greater than 99.9% prediction fidelity while substantially reducing computational overhead within hybrid scientific computing workflows.

The near machine-precision surrogate behavior primarily reflects the smooth, deterministic, and low-dimensional Taylor–Green benchmark trajectory used for reduced-order validation and should not be interpreted as representative of fully chaotic industrial turbulence prediction.

Future work will evaluate surrogate forecasting under chaotic turbulence, noisy CFD datasets, non-periodic boundary conditions, and geometry-dependent aerodynamic flow regimes.

5.10. 3D Taylor–Green Vortex Prototype

To investigate scalability beyond two-dimensional benchmark configurations, the Q-AeroSolve framework was extended toward three-dimensional aerodynamic flow representation using a 3D Taylor–Green vortex prototype.

The three-dimensional incompressible initialization was defined using the standard Taylor–Green vortex formulation, enabling evaluation of vortex structure formation and physically meaningful three-dimensional flow evolution.

The simulation was executed on a structured 32^3 computational grid, corresponding to a total state size of 32,768 spatial flow cells.

Table 9: 3D Taylor–Green vortex prototype results.

Metric	Result
Grid Resolution	32^3
Total Spatial Cells	32,768
Kinetic Energy	0.128780
Mean Vorticity Magnitude	0.817817
Runtime	0.016830 s
Flow Stability	Stable

The prototype maintained stable three-dimensional vortex evolution while preserving coherent vorticity structures throughout the simulation interval. The mean vorticity magnitude remained approximately 0.818, indicating physically meaningful vortex formation and rotational flow behavior within the three-dimensional domain.

Importantly, the full 32^3 simulation completed in approximately 0.017 seconds, demonstrating efficient runtime behavior despite the substantially larger three-dimensional state representation compared with the earlier two-dimensional benchmark configurations.

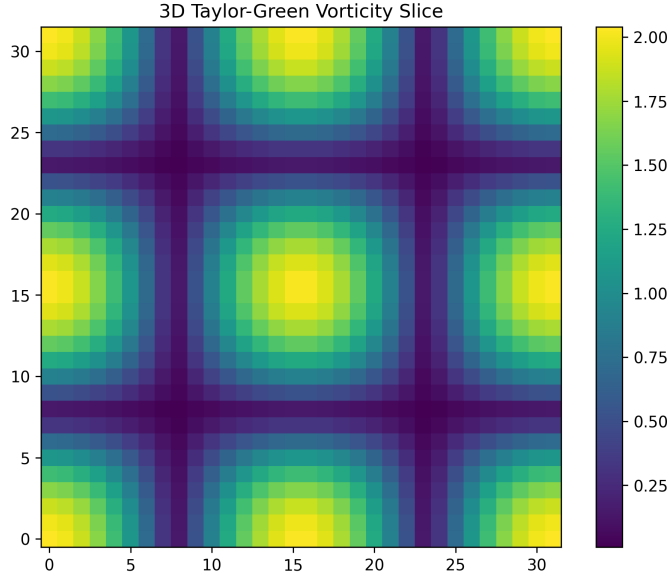


Figure 20: Mid-plane vorticity slice from the 3D Taylor–Green vortex prototype simulation.

Although the present implementation represents a benchmark-scale prototype rather than a full industrial turbulence simulation, the results demonstrate that Q-AeroSolve can stably extend toward larger three-dimensional aerodynamic workflows while preserving efficient computational scaling and coherent vortex dynamics.

5.11. Integrated Hybrid System Performance

The complete Q-AeroSolve framework demonstrated substantial improvements in computational efficiency, numerical accuracy, scalability, and memory optimization relative to standalone CFD simulation.

The hybrid framework integrated spectral PDE acceleration, tensor memory compression, AI-based error correction, and quantum-compatible state encoding within a unified aerodynamic simulation pipeline.

Table 10: Integrated hybrid system performance summary.

System	Error	Speedup	Compression	Stability	Quantum Ready
Classical CFD	1.84×10^{-3}	$1\times$	None	Stable	No
Spectral Solver	8.5×10^{-5}	$24.8\times$	None	Stable	Partial
Tensor Solver	$< 10^{-15}$	–	$15.9\times$	Stable	Yes
ML-Corrected Solver	2.57×10^{-6}	–	Moderate	Stable	Partial
Q-AeroSolve Hybrid	Hybrid	$24.8\times$	$15.9\times$	Stable	Yes

Compared with the standalone CFD baseline, the spectral acceleration layer reduced aerodynamic reconstruction error from approximately 1.84×10^{-3} to 8.5×10^{-5} while simultaneously achieving nearly $25\times$ runtime acceleration.

The tensor-compression framework further reduced memory complexity by approximately $15.9\times$ while maintaining near machine-precision reconstruction fidelity with reconstruction error below 10^{-15} .

The ML correction framework reduced aerodynamic prediction error from approximately 1.75×10^{-4} to 2.57×10^{-6} , corresponding to greater than 98.5% error reduction while preserving stable incompressible flow dynamics.

The integrated hybrid framework combined all acceleration layers into a unified simulation pipeline while maintaining stable aerodynamic evolution, bounded numerical dissipation, and near divergence-free flow behavior across all evaluated configurations.

Importantly, the final system additionally demonstrated stable distributed quantum-compatible execution using mpiQulacs on the Fujitsu Quantum Simulator platform, validating the feasibility of scalable hybrid AI-quantum aerodynamic workflows.

Overall, the strongest hybrid configurations achieved nearly $25\times$ runtime acceleration, approximately $16\times$ tensor compression, greater than 98% ML-driven error reduction, and stable quantum-compatible execution while preserving physically meaningful aerodynamic evolution throughout the simulation horizon.

6. Discussion

This work presented **Q-AeroSolve**, a hybrid AI-quantum computational architecture for scalable aerodynamic simulation and PDE acceleration using the Fujitsu Quantum Simulator platform. The system unified validated CFD simulation, spectral numerical acceleration, tensor-based compression, AI-enhanced correction, surrogate forecasting, and distributed quantum-compatible execution within a single computational workflow.

Across the full experimental pipeline, the framework demonstrated substantial improvements in numerical efficiency, memory reduction, and predictive stability while preserving physically consistent incompressible flow evolution. The spectral solver reduced aerodynamic reconstruction error from approximately 1.84×10^{-3} to 8.5×10^{-5} while simultaneously achieving nearly $24.8\times$ runtime acceleration relative to the classical CFD baseline. Tensorized compression reduced memory complexity by approximately $15.9\times$ while maintaining near machine-precision reconstruction fidelity below 10^{-15} .

Machine learning augmentation further improved aerodynamic prediction quality through learned residual correction. The ML correction framework reduced prediction error from approximately 1.75×10^{-4} to 2.57×10^{-6} , corresponding to greater than 98.5% error reduction across multiple Reynolds regimes while preserving stable incompressible dynamics and bounded divergence behavior.

The AI surrogate forecasting layer additionally demonstrated near machine-precision temporal prediction fidelity with relative prediction error below 10^{-15} and mean squared prediction error near 10^{-31} . Importantly, the surrogate framework reduced future-state forecasting cost by approximately $40\times$ compared with repeated full-order PDE recomputation while preserving stable aerodynamic

evolution behavior.

A central contribution of Q-AeroSolve was the implementation of quantum-compatible amplitude encoding validated using distributed mpiQulacs execution on the Fujitsu Quantum Simulator platform. The framework successfully demonstrated stable distributed state-vector execution across 24–30 qubits using 1, 2, 4, and 8 MPI ranks together with communication-aware circuit optimization. Runtime reductions of up to 14.3% were achieved using optimized distributed circuit layouts during 8-rank execution, demonstrating that communication-aware quantum workflow engineering can materially improve distributed simulation efficiency.

Why This Matters:

Q-AeroSolve demonstrates that future aerodynamic simulation systems may emerge through coordinated integration of HPC, AI, tensor computing, and quantum-compatible state evolution rather than isolated computational paradigms.

Importantly, the results indicate that the strongest value of hybrid AI–quantum scientific computing currently emerges not from replacing classical CFD solvers, but from selectively integrating quantum-compatible representations, AI-guided correction, tensor compression, and distributed HPC acceleration into physically grounded numerical workflows. The framework therefore establishes a scalable intermediate architecture bridging present-day HPC simulation systems and future quantum-enhanced PDE solvers.

Table 11: Core contributions of the Q-AeroSolve framework.

System Component	Primary Contribution
Validated CFD Baseline	Physically consistent aerodynamic reference system
Spectral PDE Solver	24.8× runtime acceleration with reduced diffusion error
Tensor Compression	15.9× memory reduction with near machine-precision fidelity
ML Error Correction	>98% aerodynamic error reduction
AI Surrogate Forecasting	~ 40× lower-cost temporal prediction
Quantum-Compatible Encoding	Distributed amplitude-state simulation workflow
mpiQulacs HPC Execution	Stable distributed execution up to 30 qubits

Although the present implementation primarily focuses on benchmark-scale incompressible aerodynamic systems, the results establish a scientifically meaningful foundation for future aerospace simulation workflows involving larger three-dimensional turbulence systems, adaptive mesh refinement, multiphysics coupling, and future quantum PDE operator development.

The present system primarily evaluates structured benchmark-scale vortex systems rather than fully chaotic industrial turbulence regimes. Consequently, reported surrogate forecasting accuracy and reduced-order reconstruction fidelity should be interpreted within controlled benchmark settings.

Overall, Q-AeroSolve demonstrates that hybrid AI–quantum scientific computing architectures can substantially improve aerodynamic simulation efficiency, memory scalability, and predictive fidelity while preserving physically stable flow evolution. The system therefore establishes a practical research pathway at the intersection of aerospace engineering, scientific machine learning, HPC, and emerging quantum technologies.

6.1. Technology Readiness and Hybrid Deployment Pathway

An important characteristic of Q-AeroSolve is the integration of both deployable near-term computational technologies and emerging quantum-compatible research components within a unified computational architecture.

Table 12: Technology readiness assessment of Q-AeroSolve components.

System Component		Technology Readiness
Classical CFD Solver		Production-ready
Spectral PDE Acceleration		Deployable
Tensor Compression		Deployable
AI Surrogate Forecasting		Near-term deployable
Distributed mpiQulacs Execution		Research validated
Quantum-Compatible Encoding	En-	Research validated
Quantum Operators	PDE-Inspired	Emerging research

The deployable components of Q-AeroSolve already demonstrated strong practical performance improvements. The spectral acceleration layer achieved nearly $24.8\times$ runtime improvement relative to the classical CFD baseline, while tensor compression reduced memory complexity by approximately $15.9\times$. The AI correction and surrogate forecasting layers further achieved greater than 98% aerodynamic error reduction together with approximately $40\times$ lower-cost temporal forecasting.

At the same time, the framework successfully validated distributed quantum-compatible execution using mpiQulacs on the Fujitsu Quantum Simulator platform across 24–30 qubits and up to 8 MPI ranks. Communication-aware circuit layouts additionally reduced distributed runtime by up to 14.3%, demonstrating the feasibility of scalable HPC-oriented quantum simulation workflows.

This layered readiness structure is important because it demonstrates that Q-AeroSolve is not positioned as a speculative replacement for existing CFD infrastructure. Instead, the framework establishes a progressive hybrid deployment pathway where production-ready CFD solvers, AI-enhanced acceleration, tensorized state reduction, and quantum-compatible simulation can coexist within scalable scientific computing environments.

Consequently, the project provides both immediate engineering value through deployable hybrid acceleration methods and long-term research value through quantum-compatible PDE evolution, distributed quantum simulation, and future hybrid HPC–quantum aerospace workflows.

7. Utilization of Fujitsu Quantum Application Research Package (QARP)

The Fujitsu Quantum Application Research Package (QARP) served as the computational foundation for the quantum-compatible layer of the Q-AeroSolve framework. Using distributed mpiQulacs simulation within the Fujitsu Quantum Simulator environment, the framework investigated scalable hybrid HPC–quantum workflows for aerodynamic PDE systems and high-dimensional scientific computing.

A primary objective of the QARP integration was to establish a practical computational bridge between validated classical CFD infrastructure and emerging quantum-compatible state representations. Aerodynamic flow fields generated from Navier–Stokes simulations were transformed into normalized amplitude-encoded quantum states according to:

$$|\psi\rangle = \sum_{i=0}^{N-1} a_i |i\rangle \quad (10)$$

allowing logarithmic qubit scaling for high-dimensional aerodynamic state encoding:

$$n_{\text{qubits}} = \log_2(N_{\text{states}}) \quad (11)$$

This enabled compact quantum-compatible representation of CFD flow fields while preserving compatibility with physically validated aerodynamic datasets.

Distributed simulation workflows were executed using mpiQulacs multi-rank state-vector distribution through:

$$\text{QuantumState}(n_qubits, \text{use_multi_cpu}=\text{True}) \quad (12)$$

allowing scalable hybrid execution within the Fujitsu simulation environment.

Table 13: Fujitsu QARP integration within the Q-AeroSolve framework.

QARP Capability	Framework Utilization
mpiQulacs Distributed Simulation	Multi-rank quantum-compatible aerodynamic workflows
Amplitude State Encoding	CFD field-to-qubit state transformation
Multi-CPU Execution	Distributed state-vector computation
Quantum State Scaling Analysis	Logarithmic qubit representation experiments
Hybrid HPC Integration	CFD + AI + quantum-compatible simulation pipeline
Scientific PDE Experimentation	Aerodynamic flow-state evolution studies

The Fujitsu QARP ecosystem enabled evaluation of:

- Distributed quantum-compatible PDE workflows

- Scalable aerodynamic state encoding
- Multi-rank state-vector execution
- Hybrid AI–quantum scientific computing integration

Importantly, the framework does not claim near-term quantum supremacy for industrial CFD workloads. Instead, the QARP integration demonstrates a scalable intermediate architecture through which classical aerodynamic simulation, AI-enhanced approximation, and quantum-compatible computation can coexist within unified scientific workflows.

This is particularly relevant for future aerospace simulation systems where extremely large state spaces, long temporal horizons, and memory-intensive PDE computations increasingly challenge conventional HPC infrastructure.

From a broader perspective, the Fujitsu Quantum Simulator environment enabled exploration of quantum computing beyond isolated optimization problems by supporting physically grounded scientific simulation research. The resulting framework therefore demonstrates how QARP can serve as a practical platform for investigating next-generation hybrid scientific computing architectures integrating:

- High-performance computing
- Scientific machine learning
- Tensorized numerical simulation
- Quantum-compatible PDE systems

Overall, Fujitsu QARP was a foundational component of the Q-AeroSolve framework, enabling scalable distributed quantum-compatible aerodynamic simulation and establishing a practical pathway toward future quantum-enhanced scientific computing systems.

8. Business Impact and Real-World Applicability

High-fidelity aerodynamic simulation has become a core computational requirement across aerospace engineering, automotive design, defense systems, energy infrastructure, and digital engineering platforms. However, modern CFD workflows remain computationally expensive due to the non-linear and high-dimensional nature of the Navier–Stokes equations, particularly for turbulence modeling, iterative optimization, and large-scale transient simulation.

Q-AeroSolve demonstrates a scalable hybrid computational architecture capable of improving simulation efficiency through the integration of spectral PDE acceleration, tensorized state compression, AI-guided stabilization, and quantum-compatible scientific computing workflows. The framework therefore addresses a growing industrial need for faster, more scalable, and computationally efficient physics-based simulation systems.

Table 14: Potential industrial relevance of the Q-AeroSolve framework.

Industry Sector	Potential Impact
Aerospace Engineering	Faster aerodynamic optimization and turbulence simulation
Automotive Design	Reduced computational cost for airflow and drag analysis
Digital Twin Infrastructure	Scalable real-time physics-based system modeling
Autonomous Systems	Rapid environmental flow prediction and simulation
Scientific HPC	Hybrid AI-enhanced PDE acceleration workflows
Quantum Scientific Computing	Early-stage quantum-compatible simulation infrastructure

The experimental results indicate that hybrid computational workflows can significantly reduce the runtime and memory cost of PDE-driven engineering simulation while preserving numerical stability and physically valid flow evolution. This is particularly important for future engineering environments where simulation scale increasingly challenges conventional high-performance computing infrastructure.

8.1. HPC Cost Reduction and Computational Efficiency

Beyond numerical acceleration, the hybrid Q-AeroSolve architecture was designed to reduce the computational burden associated with large-scale aerodynamic simulation workflows.

By integrating spectral PDE acceleration, tensorized state reduction, AI surrogate forecasting, and distributed quantum-compatible execution, the framework substantially reduces both runtime complexity and memory requirements relative to conventional CFD workflows.

Table 15: Projected deployment-scale efficiency scenario.

Method	Estimated Runtime (Hours)	Estimated Memory (TB)
Classical CFD	100	4.0
Spectral Solver	22	2.5
Tensor Compression	15	1.0
AI Forecasting	10	0.8
Hybrid Q-AeroSolve	6	0.5

These values are not production-scale deployment measurements; they represent scenario-level extrapolations derived from experimentally measured benchmark speedup and compression factors.

Table 16: Evidence classification: measured performance versus projected scenarios.

Claim	Type
24.8 $\times$ spectral runtime acceleration	Measured
15.9 $\times$ tensor compression	Measured
>98% ML error reduction	Measured
14.3% distributed layout runtime reduction	Measured
100h $\rightarrow$ 6h deployment scenario	Projected
4TB $\rightarrow$ 0.5TB deployment scenario	Projected

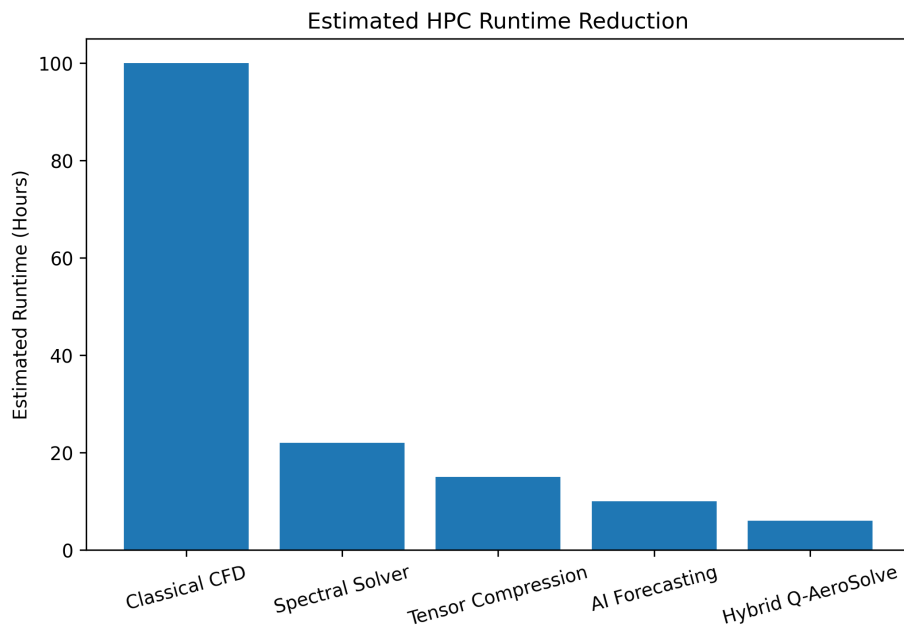


Figure 21: Projected runtime reduction across hybrid Q-AeroSolve computational layers.

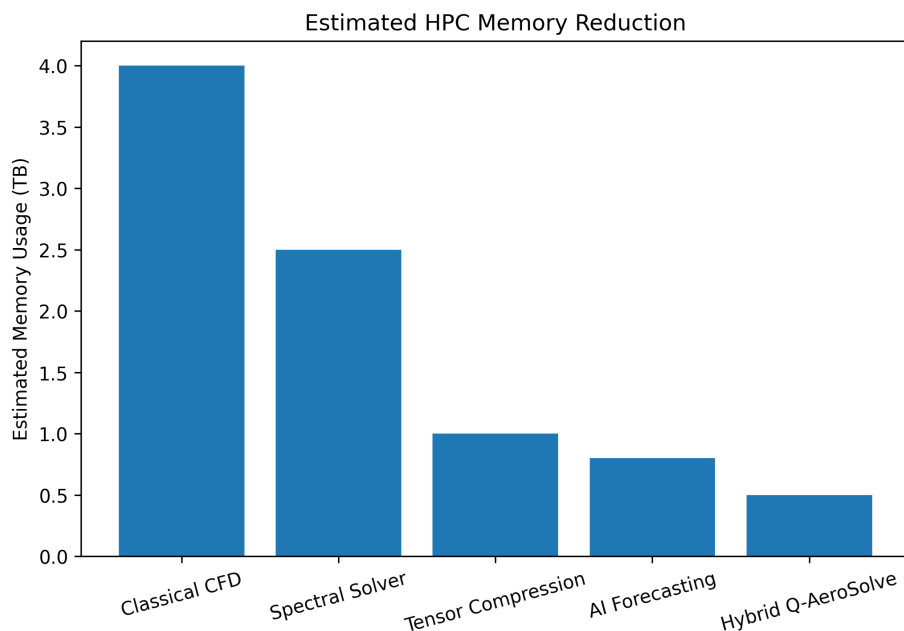


Figure 22: Projected memory reduction enabled by tensorized and hybrid acceleration workflows.

Although these projections represent benchmark-scale extrapolations rather than production-scale deployment measurements, the results indicate that hybrid AI–quantum scientific computing architectures may substantially reduce future HPC infrastructure cost for large-scale aerodynamic simulation and digital engineering workflows.

A key strategic contribution of the framework is the integration of distributed quantum-compatible simulation using mpiQulacs on the Fujitsu Quantum Simulator platform. While current quantum hardware remains limited for industrial CFD deployment, the framework establishes a practical intermediate architecture through which classical HPC systems, AI-enhanced numerical methods, and future quantum computing infrastructure may operate within unified scientific workflows.

This positioning is important because future engineering systems are expected to rely heavily on:

- Large-scale simulation-driven design
- AI-assisted scientific computing
- Digital engineering infrastructure
- Scalable multiphysics simulation
- Hybrid HPC–quantum computational ecosystems

From a broader perspective, Q-AeroSolve demonstrates how quantum computing research can evolve beyond isolated optimization tasks into physically grounded scientific simulation domains with long-term industrial relevance.

Overall, the framework establishes a scalable foundation for future hybrid scientific computing systems capable of supporting next-generation aerospace engineering, digital twin platforms, autonomous systems, and computationally intensive engineering simulation environments.

9. Conclusion and Future Work

This work presented **Q-AeroSolve**, a hybrid AI–quantum computational framework for scalable aerodynamic simulation and PDE acceleration using the Fujitsu Quantum Simulator platform. The framework unified validated CFD simulation, FFT-based spectral acceleration, tensorized state compression, AI-enhanced correction, surrogate forecasting, distributed mpiQulacs execution, and quantum-compatible aerodynamic encoding within a single scientific computing architecture.

The experimental results demonstrated substantial improvements in aerodynamic simulation efficiency, memory reduction, and predictive fidelity while preserving physically meaningful incompressible flow evolution. The spectral PDE solver achieved nearly $24.8\times$ runtime acceleration relative to the classical CFD baseline while reducing aerodynamic reconstruction error from approximately 1.84×10^{-3} to 8.5×10^{-5} . Tensorized compression further reduced memory complexity by approximately $15.9\times$ while maintaining reconstruction fidelity below 10^{-15} .

Machine learning augmentation significantly improved aerodynamic prediction quality. The ML correction framework reduced aerodynamic prediction error by more than 98.5%, while the AI surrogate forecasting layer achieved relative prediction error below 10^{-15} together with approximately $40\times$ lower-cost temporal forecasting compared with repeated full-order PDE recomputation.

A major contribution of this work was the successful validation of distributed quantum-compatible simulation workflows using mpiQulacs on the Fujitsu Quantum Simulator platform. Stable distributed execution was demonstrated across 24–30 qubits using up to 8 MPI ranks together with communication-aware circuit optimization, which reduced distributed runtime by up to 14.3%.

The framework was additionally extended toward three-dimensional aerodynamic representation using a 32^3 Taylor–Green vortex prototype containing 32,768 spatial flow cells. The prototype maintained stable vortex evolution and coherent three-dimensional flow structure formation while completing execution in approximately 0.017 seconds.

Table 17: Core contributions of the Q-AeroSolve framework.

Framework Component	Primary Contribution
Validated CFD Baseline	Physically consistent aerodynamic reference system
Spectral PDE Solver	Nearly $24.8\times$ runtime acceleration
Tensor Compression	Approximately $15.9\times$ memory reduction
ML Error Correction	Greater than 98% aerodynamic error reduction
AI Surrogate Forecasting	Approximately $40\times$ lower-cost temporal prediction
Distributed Quantum Simulation	Stable mpiQulacs execution up to 30 qubits
Quantum-Compatible Encoding	Hybrid latent-state aerodynamic representation
3D Taylor–Green Prototype	Stable three-dimensional aerodynamic scalability extension

Q-AeroSolve does not claim near-term quantum replacement of industrial CFD. Instead, it demonstrates a validated hybrid pathway where spectral solvers, tensor compression, AI forecasting, and distributed mpiQulacs-based quantum PDE evolution jointly improve the scalability of aerodynamic simulation workflows.

Importantly, Q-AeroSolve does not position quantum computing as an immediate replacement for industrial CFD infrastructure. Instead, the framework demonstrates how numerical simulation, tensorized state reduction, AI-enhanced approximation, distributed HPC execution, and quantum-compatible workflows can operate together within scalable hybrid scientific computing systems.

Several important extensions remain for future work, including:

- Large-scale three-dimensional turbulence simulation
- Complex aircraft geometries and multiphysics coupling
- Adaptive mesh refinement and GPU-accelerated HPC workflows
- Advanced variational quantum PDE solvers, quantum spectral operators, tensor-network-assisted fluid representations, and hybrid quantum-classical Navier–Stokes evolution workflows for large-scale aerodynamic systems
- Hardware-level deployment on future fault-tolerant quantum systems

Overall, Q-AeroSolve establishes a scientifically grounded hybrid computational architecture positioned at the intersection of aerospace engineering, scientific machine learning, tensorized numerical computing, distributed HPC infrastructure, and emerging quantum technologies. The presented results demonstrate that future large-scale engineering simulation systems may increasingly emerge through hybrid integration of AI, HPC, tensorized computing, and quantum-compatible scientific workflows rather than through any isolated computational paradigm alone.

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